

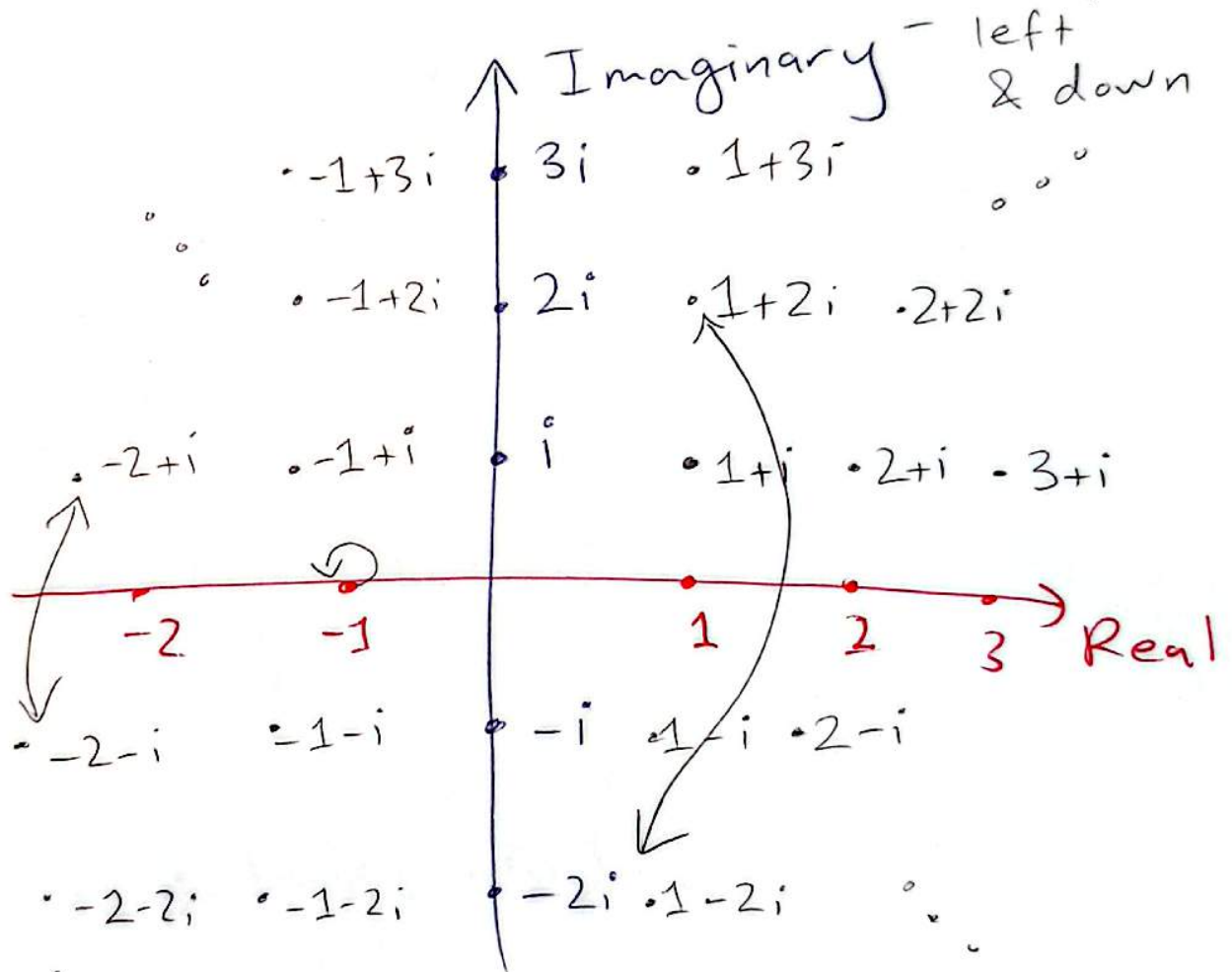
Complex Numbers

The complex number system is an extension of the real number system containing square roots of negative numbers.

$$i^2 = -1$$

+ right
& up

- left
& down



$a+bi$ a, b real numbers

real part imaginary part

What are the solutions to:

$$x^2 = -1 \rightarrow x = i \text{ OR } -i$$

Why? $(-i)^2 = [(-1) \cdot i]^2$
 $= (-1)^2 \cdot i^2$
 $= 1 \cdot (-1) = -1$

$$x^2 = -9 \rightarrow$$

$$x = 3i \text{ OR } -3i$$

$$(3i)^2 = 3^2 i^2 = 9 \cdot (-1) = -9$$

$$x^2 = -5 \rightarrow$$

$$x = \sqrt{5} \cdot i \text{ OR } -\sqrt{5} \cdot i$$

$$= i\sqrt{5} \text{ OR } -i\sqrt{5}$$

Complex Algebra

$$\text{e.g. } (3+4i) + (2+5i) = ?$$

$$= (3+2) + (4i+5i)$$

$$= \boxed{5+9i}$$

$$\text{e.g. } (3+4i) - (2+5i)$$

$$= (3-2) + (4i-5i)$$

$$= \boxed{1-i}$$

$$g. (3+4i)(2+5i)$$

$$3 \cdot 2 + 3 \cdot 5i + 4i \cdot 2 + (4i) \cdot (5i)$$

$$= 6 + 15i + 8i + 20 \underbrace{i \cdot i}_{i^2 = -1}$$

$$= 6 + 23i + 20(-1)$$

$$= (6-20) + 23i = \boxed{-14 + 23i}$$

Powers of i go in a cyclic pattern

i^1	i^2	i^3	i^4
"	"	"	"
i	-1	$-i$	1

i^5	i^6	i^7	i^8
"	"	"	"
$i^4 \cdot i^1$	$i^4 \cdot i^2$	$i^4 \cdot i^3$	$i^4 \cdot i^4$
$= 1 \cdot i$	$= 1 \cdot (-1)$	$= 1 \cdot (-i)$	$= 1 \cdot 1$
"	"	"	"
i	(-1)	$-i$	1

g. What is i^{51} ?

$$\begin{aligned} i^{51} &= i^{48+3} = i^{48} \cdot i^3 \\ &= (i^4)^{12} \cdot i^3 \\ &= 1 \cdot i^3 \\ &= \boxed{-i} \end{aligned}$$

Complex Conjugates and Rationalizations

If $z = a+bi$, $\bar{z} = a-bi$

e.g. conjugates of: $5+7i \leftrightarrow 5-7i$

$$-1-2i \leftrightarrow -1+2i$$

$$3 = 3+0i \leftrightarrow 3-0i = 3$$

$$3i = 0+3i \leftrightarrow 0-3i = -3i$$

conjugate of conjugate is original
number $\overline{(\bar{z})} = z$

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special property of conjugates

$$(a+bi)(a-bi) = a^2 + b^2$$

$\downarrow$
 $z \cdot \bar{z} = \text{length squared}$
 $\text{of } z$

$\downarrow$
positive real number
as long as $z = a+bi$
is not 0

Using conjugates to solve complex
quotients

e.g. $7-i$ $\rightarrow$ Need to express as
 $a+bi$, a & b are real

$3-5i$

$z = 3-5i$, $\bar{z} = ?$

$\bar{z} = 3+5i$

$$\frac{(7-i) \cdot (3+5i)}{(3-5i) \cdot (3+5i)} = \frac{7 \cdot 3 + 7 \cdot 5i - i \cdot 3 - i \cdot (5i)}{3^2 + (-5)^2}$$

$$= \frac{21 + 35i - 3i + 5}{9 + 25}$$

$$= \frac{\underline{21} + \cancel{35i} - \cancel{3i} + \underline{5}}{9 + 25}$$

$$= \frac{26 + 32i}{34} = \frac{26}{34} + \frac{32i}{34}$$

$$= \left[\frac{13}{17} + \frac{16}{17}i \right]$$

$$\frac{7-i}{3-5i} = \frac{13}{17} + \frac{16}{17}i$$

Real Part Imag. Part

$$a = \frac{13}{17}, \quad b = \frac{16}{17}$$

Quadratic Equations w/ Complex Roots

e.g. $5x^2 + 2x + 1 = 0$

$\downarrow$ $\downarrow$ $\downarrow$
 $a=5$ $b=2$ $c=1$

$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$

discriminant

$$x = \frac{-2 \pm \sqrt{2^2 - 4(5)(1)}}{2(5)}$$

$$= \frac{-2 \pm \sqrt{4 - 20}}{10}$$

$$= \frac{-2 \pm \sqrt{-16}}{10}$$

$$= \frac{-2 \pm 4i}{10} = \frac{2(-1 \pm 2i)}{2 \cdot 5}$$

$$= \frac{-1 \pm 2i}{5}$$